



World Heat Flow Portal (WHFP) project

Phase Two (2025-2028)

The World Heat Flow Portal (WHFP) project, a continuation of the [World Heat Flow Database](#) (2022–2025), is dedicated to consolidating and enhancing the current research data infrastructure for global terrestrial heat-flow data (www.heatflow.world). This project is designed to provide the geoscientific community with efficient and open access to and utilization of over a century's worth of research and data collection. The project emphasizes the improvement and expansion of the current research data infrastructure and web-based data portal, focusing on increased resilience, interoperability, and user experience. Key initiatives include

- concluding the global data assessment
- transitioning the data publication process from the current prototype to a fully semi-automated system,
- expanding the database with new data types and enhanced metadata standards (esp. for marine heat flow data), and
- strengthening both front-end and back-end functionalities.

These enhancements will include additional API representations, advanced analytical tools, educational resources, and community-driven data and metadata enrichment. A crucial aspect of the WHFP is the integration of marine heat-flow data, bridging the fields of geoscience and oceanography, and establishing standard practices across these disciplines.

The project (phase two: 09.2025-08.2028) is funded by the "The [Deutsche Forschungsgemeinschaft](#)" (DFG, German Research Foundation). See the [GEPRIS project overview](#).

Project partners are the

- GFZ Helmholtz Centre for Geosciences (Sven Fuchs, Kirsten Elger, Ben Norden, Sam Jennings, Samah Elbarbary, Viktoria Dergunova)
- Technical University of Dresden (Stephan Mäs, Nikolas Ott)
- University of Bremen/MARUM (Achim Kopf, Florian Neumann).

[Contact](#) the project PI. The full team behind the development of phase two is listed [here](#). External partners are listed [here](#):

Upcoming events

- Jun 2026 - Presentation at World Geothermal Congress WGC2026 (Calgary, Canada)
- Mar 2026 - 2nd project workshop (Dresden)
- Jan 2026 - 1st project workshop (8-9 January, Bremen)

Project record

- Oct 2025 - European Geothermal Congress EGC2025 (6-9 Oct, Zurich, Switzerland) [[presentation](#)]
- Sep 2025 - Kick-off meeting (22-23 Sep, Potsdam)
- Sep 2025 - Stakeholder Meeting during IASPEI/IGA Meeting (Portugal, 31 Aug - 5 Sep)
- July-Aug - Open online evaluation phase of heatflow.world

Previous activities

The prototype of this research data infrastructure was developed between 2022 and 2025, during phase one of the project (World Heat Flow Database Project). Check out the track record of phase one [here](#).

- Nov 2025 - 1st project workshop (2-day kick-off workshop, Potsdam)

Publications and Outreach

Conferences and presentations

- Sven Fuchs, Florian Neumann, Elif Pazvantoğlu, Samah Elbarbary, Alexej Petrunin, Ben Norden (2025). Heat Flow in Europe: A new quality-controlled resource for geothermal energy exploration. **European Geothermal Congress EGC2025** (Zurich, Switzerland) - (ORAL).
- Samah Elbarbary, Florian Neumann, Elif Balkan-Pazvantoğlu, Alexej Petrunin, Ben Norden, Sven Fuchs (2025). A new Quality-Controlled Heat Flow Database for Europe. **IGA/IASPEI Joint Scientific Meeting 2025** (Lisbon, Portugal) - (POSTER)
- Ben Norden, Sven Fuchs, Eskil Salis Gross (2025) Establishing a New Borehole Thermal Data Scheme for Climate Studies. **IGA/IASPEI Joint Scientific Meeting 2025** (Lisbon, Portugal) - (POSTER)
- Sven Fuchs (2025). A Global Heat Flow Data Revolution: Standardization, Quality Assessment, and Future Perspectives. **IGA/IASPEI Joint Scientific Meeting 2025** (Lisbon, Portugal) - (ORAL)
- M. Verdoya, G. Beardsmore, S. Fuchs, S. Liu, R. Harris, R. Negrete Aranda (2025). The lithospheric heat flow global data assessment project: state of the art and developments. **IGA/IASPEI Joint Scientific Meeting 2025** (Lisbon, Portugal) - (ORAL).

Funding & Team

- The project (phase two: 09.2025-08.2028) is funded by "The [Deutsche Forschungsgemeinschaft](#)" (DFG, German Research Foundation). Visit the [DEPRIS entry](#).
- The project PI's are: [Dr. Kirsten Elger](#) (GFZ), [Dr. Sven Fuchs](#) (GFZ), [Professor Dr. Achim Kopf](#) (MARUM), [Dr.-Ing. Stephan Mäs](#) (TU Dresden)
- Please visit the [detailed list of project staff](#).

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